

Python Data Analysis Internship Project

Manufacturing Production Analytics

Business Scenario

Prime Manufacturing Ltd. manufactures household electrical appliances across multiple factories and production lines. The company has experienced steady growth, but management has identified inconsistencies in production output, increasing machine downtime, varying product quality and rising operational costs. You have recently joined the Manufacturing Analytics team as a Junior Manufacturing Data Analyst. Your responsibility is to analyse the production dataset and present meaningful findings that will support operational decision-making and continuous improvement.

Dataset

The dataset contains production activities collected across all factories over a 12-month period.

Dataset Columns
Production ID
Production Date
Factory
Production Line
Shift
Machine ID
Machine Type
Operator ID
Product Category
Planned Production
Actual Production
Defective Units
Downtime (Minutes)
Production Hours
Energy Consumption (kWh)
Maintenance Cost (\$)
Unit Production Cost (\$)
Temperature (°C)

Humidity (%)
Quality Inspection Score

Project Requirements

Task 1 – Data Exploration

Perform an initial exploration of the dataset to understand its structure and contents. Your analysis should include all relevant information necessary before data cleaning and analysis begin.

Task 2 – Data Cleaning

Perform all necessary data cleaning operations required to prepare the dataset for analysis. Document every cleaning step performed.

Task 3 – Exploratory Data Analysis

Perform a detailed exploratory data analysis of the manufacturing operations covering production performance, factory performance, production lines, machine performance, product quality, operational efficiency, downtime, energy consumption, maintenance and workforce performance.

Task 4 – Grouping and Aggregation

Perform meaningful grouping and aggregation to compare business performance across operational areas and identify important trends.

Task 5 – Data Visualisation

Create appropriate visualisations

Task 6 – Management Report

Prepare a professional report summarising findings, concerns, opportunities for improvement and recommendations supported by data.

Deliverables

- Python Notebook (.ipynb)
- Cleaned Dataset (.csv)
- Management Report (.pdf)

Important Instructions

- Perform all analysis using Python.
- Organise the notebook with clear Markdown headings and explanations.
- Use meaningful variable names and good coding practices.
- Include titles and labels on all visualisations.
- Support conclusions with evidence from the analysis.
- Ensure the notebook runs from start to finish without errors.
- Submit all required deliverables before the deadline.